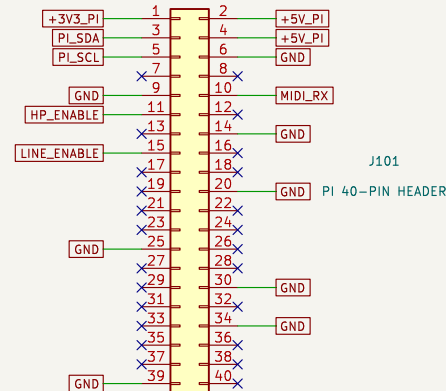


ARDOR / CODEC ZERO I/O

01 / System interface and power

REV A | ENGINEERING PROTOTYPE – VALIDATE BEFORE PRODUCTION

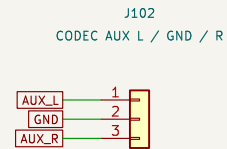
PI HEADER / STACKING INTERFACE



```
GPI015 / pin 10: UART RX, 31250 baud, 8-N-1
GPI017 / pin 11: headphone enable, active high
GPI027 / pin 13: reserved for Codec Zero button
```

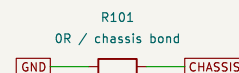
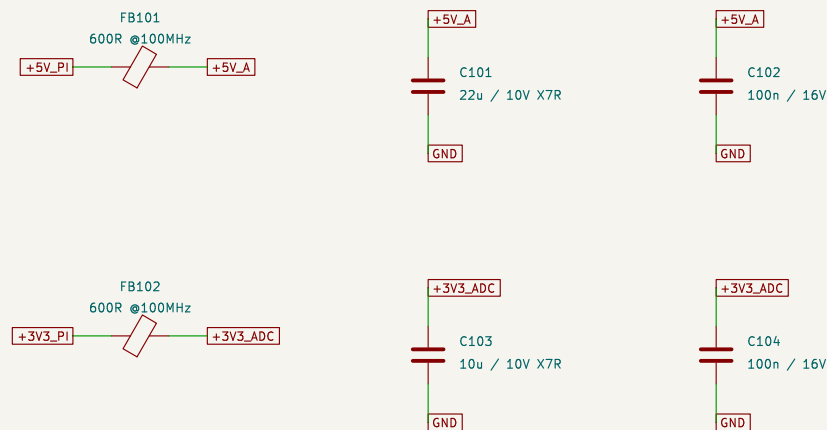
GPIO22 / pin 15: line relay enable, active high
I2C1: GPIO2/3. ADS1115 address 0x48.
Reserved Codec pins: GPIO18–21 I2S; GPIO23/24 LEDs;
GPIO27 button; GPIO0/1 HAT EEPROM. No added loads.

AUX HEADER HARNESS



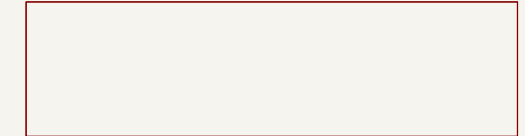
J102 pin order is OUR harness definition.
Map to the actual Codec Zero L/R/GND pads.
Never connect the mono speaker BTL output here.

FILTERED BRANCHES / 5 V REGULATED INPUT ONLY



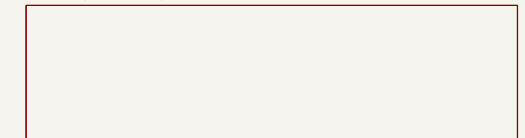
Bond enclosure at connector entry. Short, wide copper. Use one continuous PCB ground plane; place by function. Do not route ESD or charge-pump return through ADC/audio. 5 V source: 4.75–5.25 V; reserve 150 mA beyond Pi/Codec. Pi provides 3.3 V. No second supply or USB back-feed.

02 / Isolated MIDI input



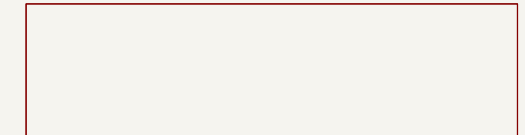
File: 02_MIDI.kicad_sch

03 / Expression input



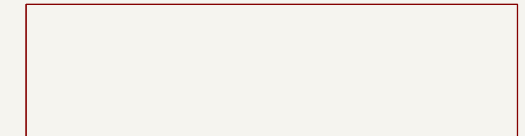
File: 03_Expression.kicad_sch

04 / Audio distribution



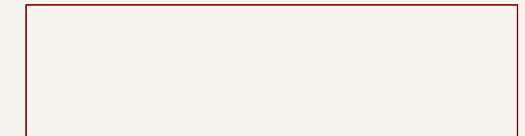
File: 04_Audio.kicad_sch

05 / Line and amp feed



File: 05_Line_Amp.kicad_sch

06 / Stereo headphones



File: 06_Headphones.kicad_sch

Prototype design; see DESIGN_NOTES.md for release checks

ARDOR – Codec Zero expansion

Sheet: /
File: Ardor_IO.kicad_sch

Title: 01 / System interface and power

Size: A3

Date: 2026-09-05

Rev: A

KiCad E.D.A. 9.0.5

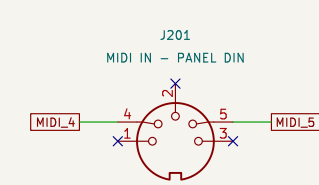
Id: 1/6

ARDOR / CODEC ZERO I/O

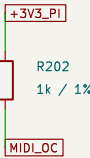
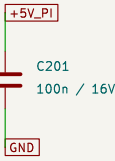
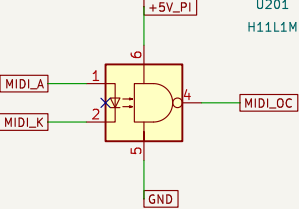
02 / Isolated MIDI input

REV A | ENGINEERING PROTOTYPE – VALIDATE BEFORE PRODUCTION

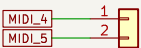
5–PIN DIN / FLOATING CURRENT LOOP



Female DIN, 180 degrees. Use contact numbers, not an assumed solder–side drawing. Pins 1, 2, 3 have NO PCB connection. Metal shell bonded to enclosure mechanically.

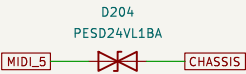
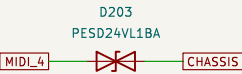
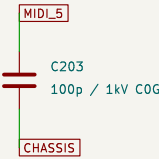
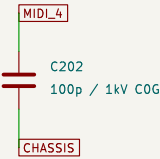


J202 TO MIDI PANEL / 4,5



MIDI LED on -> UART low. No inverter required. Open collector pulled up ONLY to 3.3 V. Disable serial console; enable stable UART RX. Never use a 5 V pull–up on Raspberry Pi GPIO.

CONNECTOR ESD / RF NETWORK



D202 clamps differential transients; D203/D204 divert common–mode ESD to CHASSIS (24 V standoff). Keep the MIDI loop copper isolated from logic GND: >= 3 mm clearance around U201 input and nets. Common–mode range is limited by D203/D204. Functional ground–loop isolation; NOT safety isolation. No MIDI–pin connection to logic supply. Place TVS and 100p RF capacitors at DIN harness entry. Qualify IEC 61000–4–2 on the assembled enclosure, including common–mode strikes and UART recovery.

Prototype design; see DESIGN_NOTES.md for release checks

ARDOR – Codec Zero expansion

Sheet: /02 / Isolated MIDI input/
File: 02_MIDI.kicad_sch

Title: 02 / Isolated MIDI input

Size: A3
KiCad E.D.A. 9.0.5

Date: 2026–09–05

Rev: A

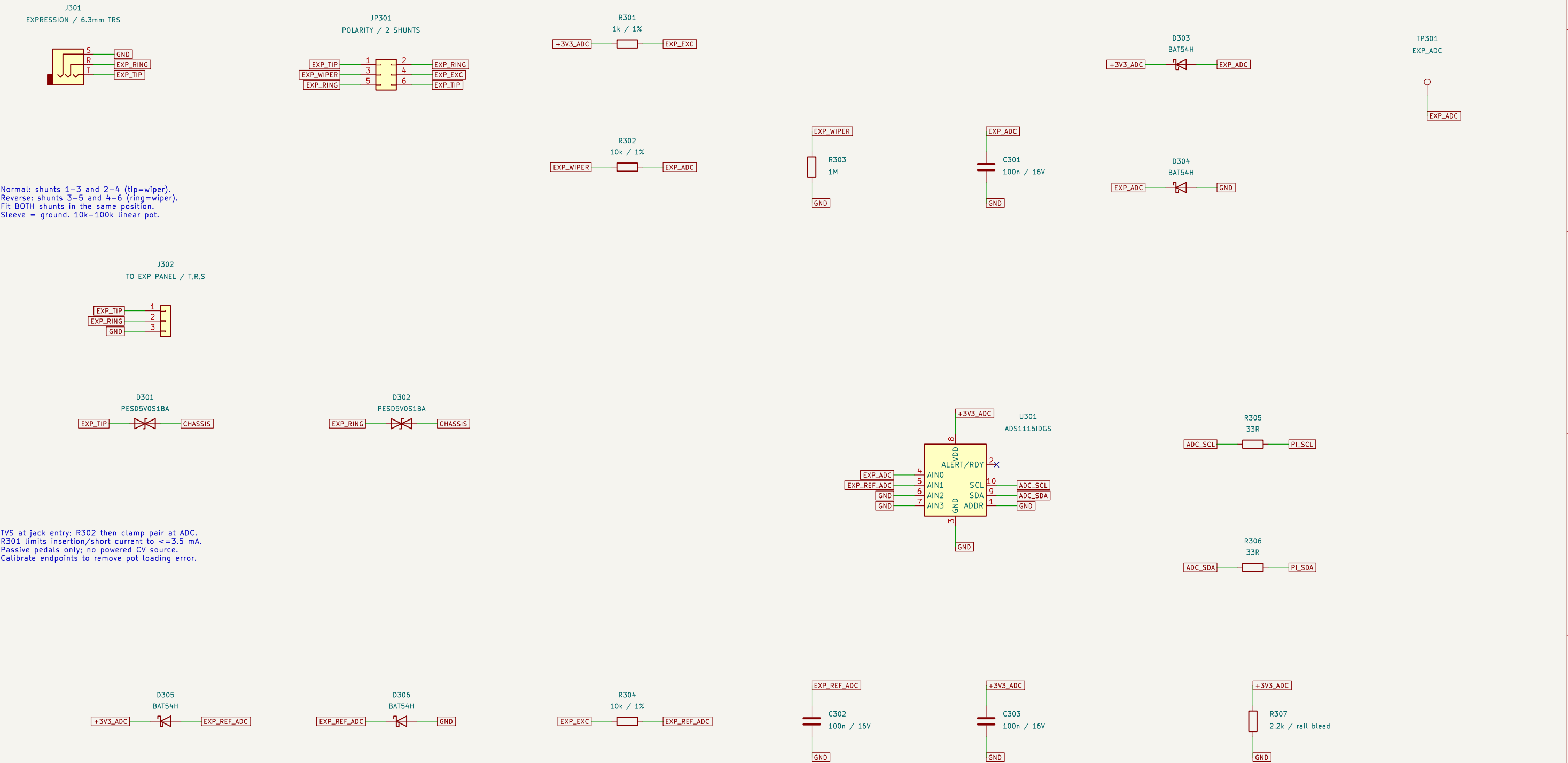
Id: 2/6

ARDOR / CODEC ZERO I/O

03 / Expression input

REV A | ENGINEERING PROTOTYPE – VALIDATE BEFORE PRODUCTION

PASSIVE TRS PEDAL / TIP–RING SELECT



I2C address 0x48; internal reference; PGA +/-4.096 V; 128 SPS. Read AIN0 and AIN1 single-ended.
Use AIN0 / AIN1 ratio, endpoint calibration, smoothing and deadband. No added I2C pull-ups (Pi has them).

Prototype design; see DESIGN_NOTES.md for release checks

ARDOR – Codec Zero expansion

Sheet: /03 / Expression input/

File: 03_Expression.kicad_sch

Title: 03 / Expression input

Size: A3

Date: 2026-09-05

Rev: A

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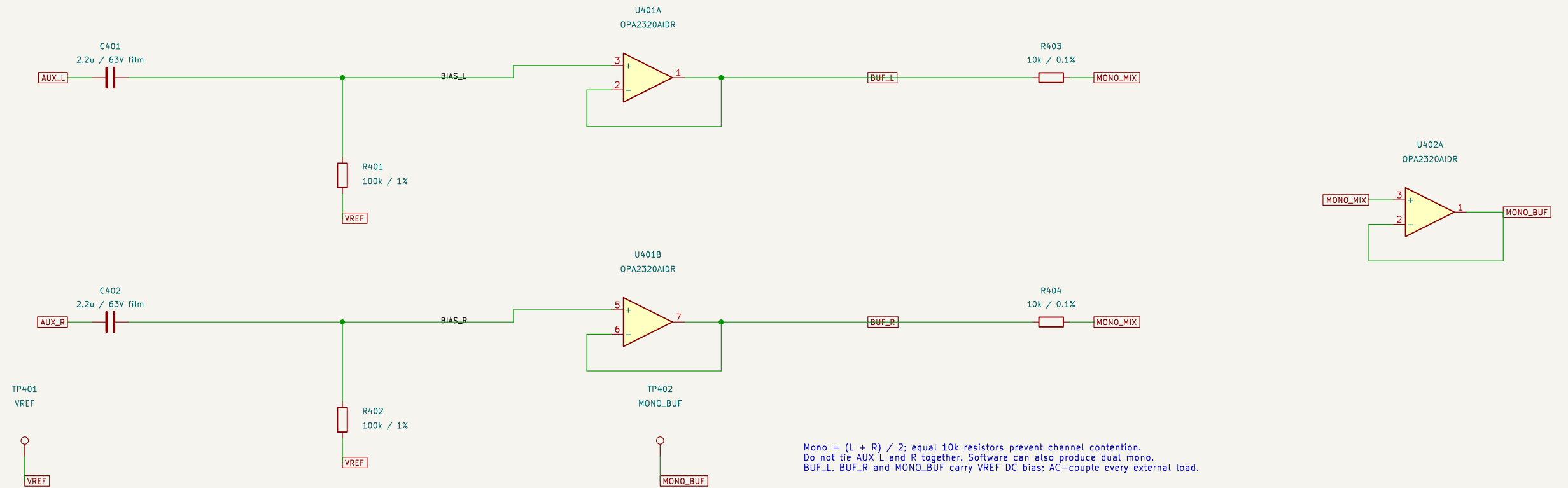
Id: 3/6

ARDOR / CODEC ZERO I/O

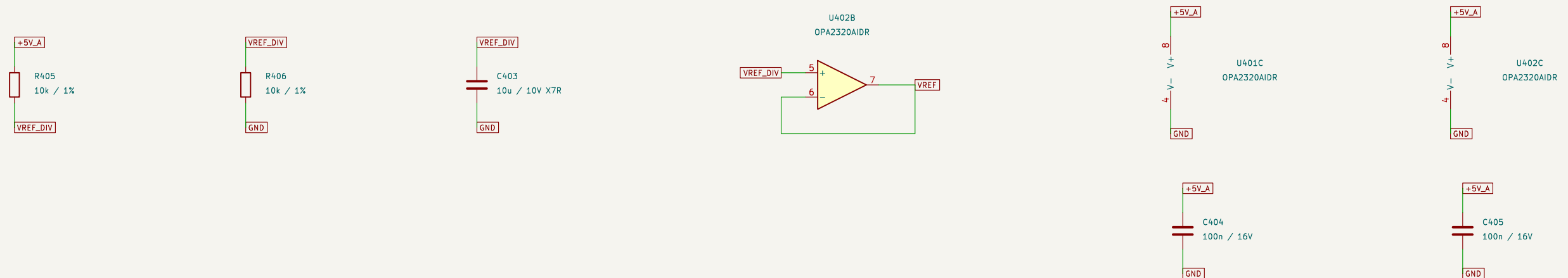
04 / Audio distribution

REV A | ENGINEERING PROTOTYPE – VALIDATE BEFORE PRODUCTION

AC-COUPLED STEREO INPUT / UNITY BUFFERS



MIDRAIL / LOCAL BYPASS



Nominal audio target ≤ 1.0 Vrms per AUX channel. Unity gain throughout.
OPA2320: RRIO on 5 V, unity stable; 2.5 V bias permits bipolar audio swing.
2.2u/100k input pole ~ 0.72 Hz. Keep VREF local; no large capacitor directly on buffer output.

Prototype design; see DESIGN_NOTES.md for release checks

ARDOR – Codec Zero expansion

Sheet: /04 / Audio distribution/
File: 04_Audio.kicad_sch

Title: 04 / Audio distribution

Size: A3

Date: 2026-09-05

Rev: A

Size: A5	
KiCad E.D.A. 9.0.5	

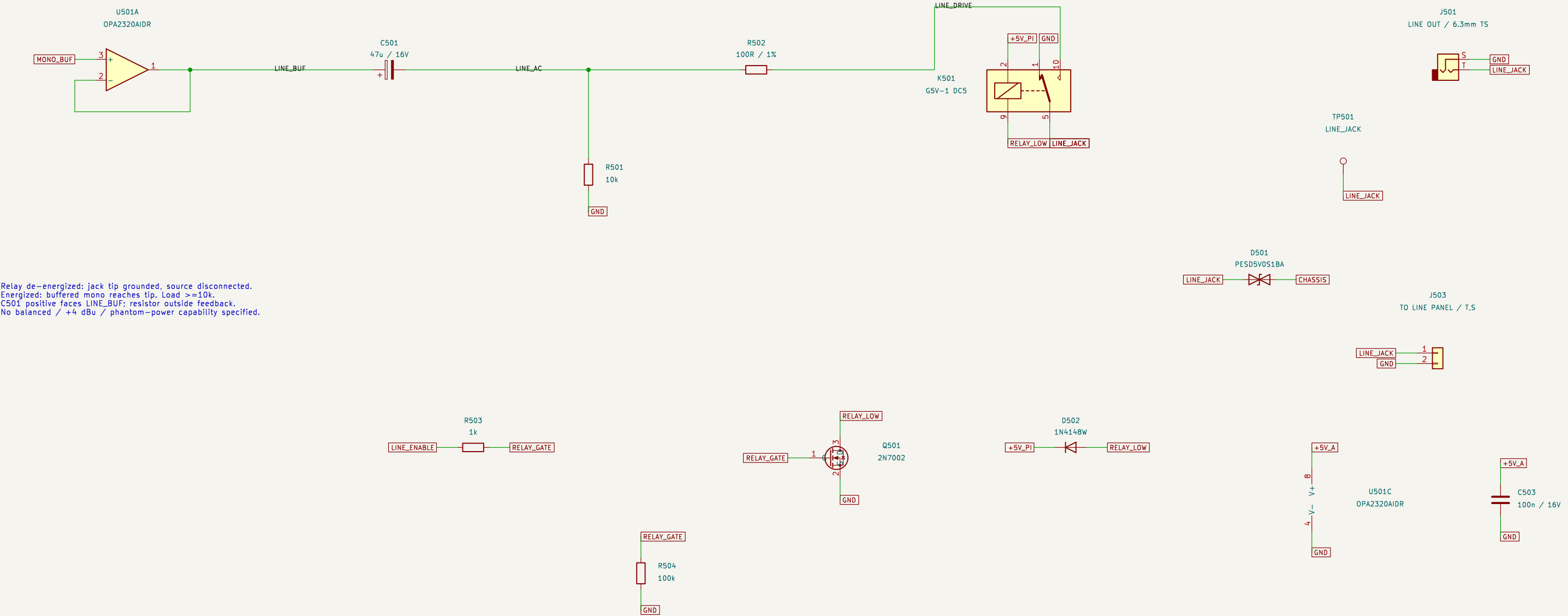
id: 4/6

ARDOR / CODEC ZERO I/O

05 / Line and amp feed

REV A | ENGINEERING PROTOTYPE – VALIDATE BEFORE PRODUCTION

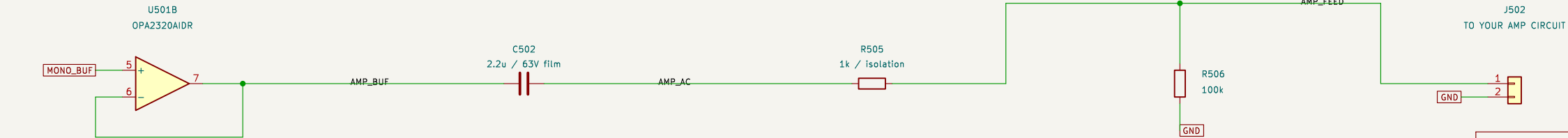
MONO LINE / DC BLOCK / STARTUP RELAY MUTE



Relay de-energized: jack tip grounded, source disconnected.
Energized: buffered mono reaches tip. Load >=10k.
C501 positive faces LINE_BUF; resistor outside feedback.
No balanced / +4 dBu / phantom-power capability specified.

Amp input >=100k. Add your gain, output connector,
external-port protection and mute in that project.
J502 is internal, NOT a speaker / amp output jack.

AMP CIRCUIT PLACEHOLDER / INTERNAL HARNESS



Prototype design; see DESIGN_NOTES.md for release checks

ARDOR – Codec Zero expansion

Sheet: /05 / Line and amp feed/

File: 05_Line_Amp.kicad_sch

Title: 05 / Line and amp feed

Size: A3 Date: 2026-09-05

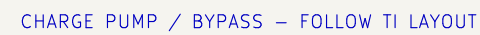
KiCad E.D.A. 9.0.5

Rev: A

Id: 5/6

06 / Stereo headphones

STEREO DIRECTPATH HEADPHONE DRIVER / -6 dB



KiCad E.D.A. 9.0.5